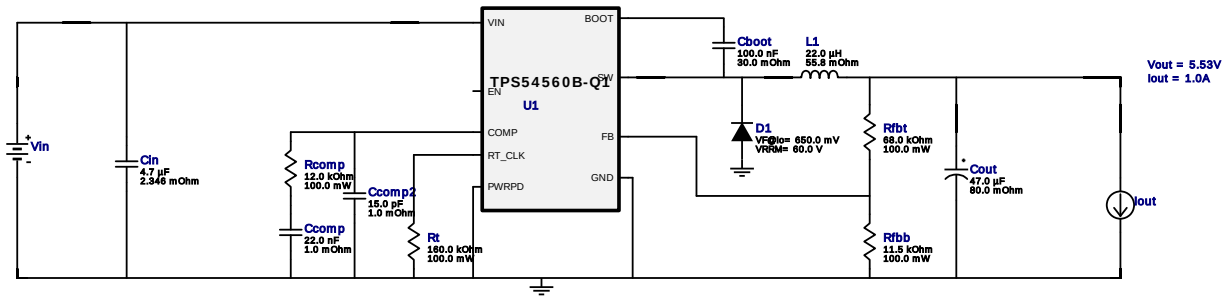
WEBENCH ® Design Report

VinMin = 9.0V
VinMax = 14.4V
Vout = 5.53V
Iout = 1.0A

Device = TPS54560BQDDARQ1
Topology = Buck
Created = 2025-01-27 13:54:27.043
BOM Cost = $3.83
BOM Count = 12
Total Pd = 0.35W

Design : 1 TPS54560BQDDARQ1
TPS54560BQDDARQ1 10V-14V to 5.33V @ 1A



1. This regulator device is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application.View WEBENCH(R) Disclaimer.

Design Alerts

Component Selection Information

The TPS54560B-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application.

Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM188R71C104KA01D Series= X7R	Cap= 100.0 nF ESR= 30.0 mOhm VDC= 16.0 V IRMS= 1.7 A	1	$0.01	0603 5 mm²
Ccomp	MuRata	GRM188R71H223KA01D Series= X7R	Cap= 22.0 nF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
Ccomp2	MuRata	GRM1885C1H150JA01D Series= C0G/NP0	Cap= 15.0 pF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	$0.02	0603 5 mm²
Cin	TDK	C2012X7R1V475K125AC Series= X7R	Cap= 4.7 uF ESR= 2.346 mOhm VDC= 35.0 V IRMS= 4.2602 A	1	$0.15	0805 7 mm²
Cout	AVX	TPSD476K016R0080 Series= TPS	Cap= 47.0 uF ESR= 80.0 mOhm VDC= 16.0 V IRMS= 1.232 A	1	$0.58	7343-31 59 mm²
D1	Fairchild Semiconductor	FSV360FP	VF@Io= 650.0 mV VRRM= 60.0 V	1	$0.13	SOD-123HE 13 mm²
L1	Bourns	SRU1038-220Y	L= 22.0 µH 55.8 mOhm	1	$0.49	SRU1038 144 mm²
Rcomp	Yageo	RC0603FR-0712KL Series= ?	Res= 12.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²

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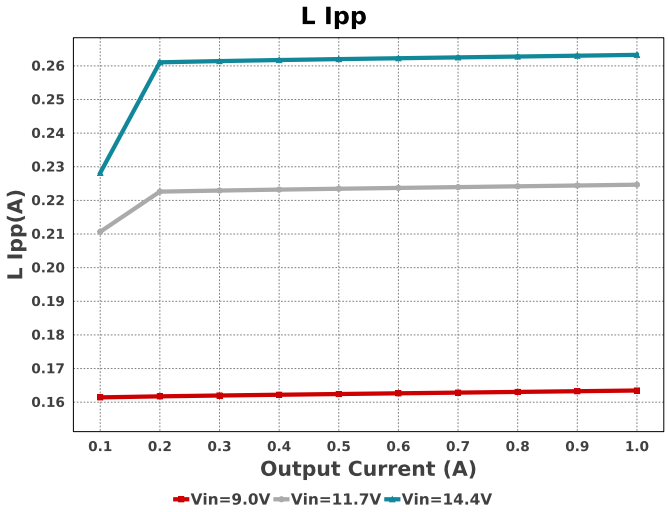
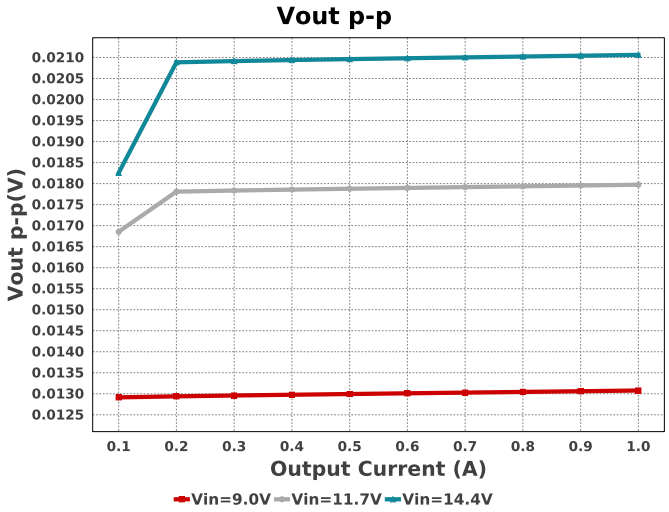
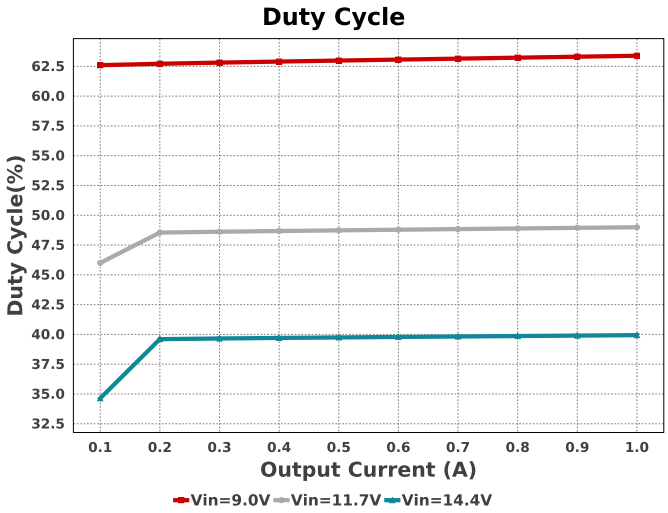
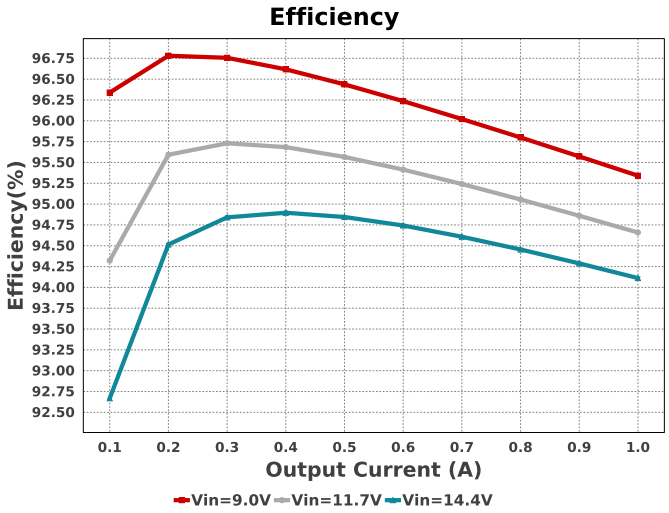
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rfbb	Yageo	RC0603FR-0711K5L Series= ?	Res= 11.5 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rfbt	Yageo	RC0603FR-0768KL Series= ?	Res= 68.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rt	Yageo	RC0603FR-07160KL Series= ?	Res= 160.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
U1	Texas Instruments	TPS54560BQDDARQ1	Switcher	1	$2.40	DDA0008E 55 mm²









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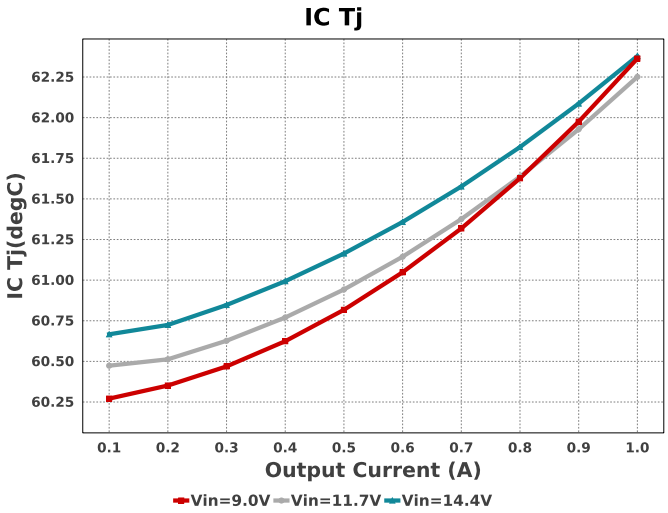

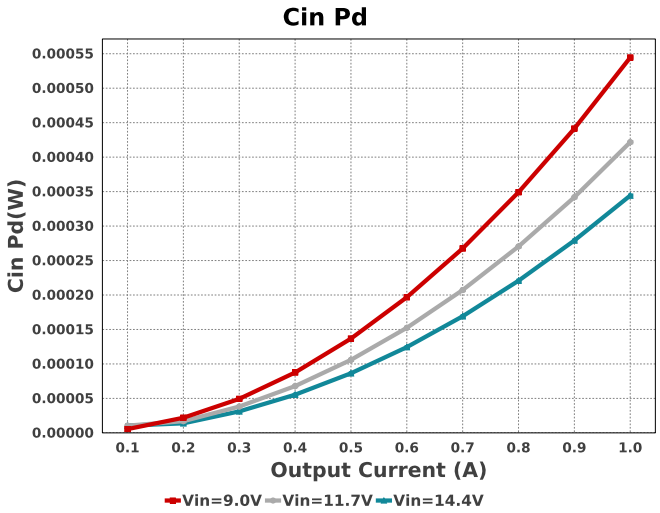

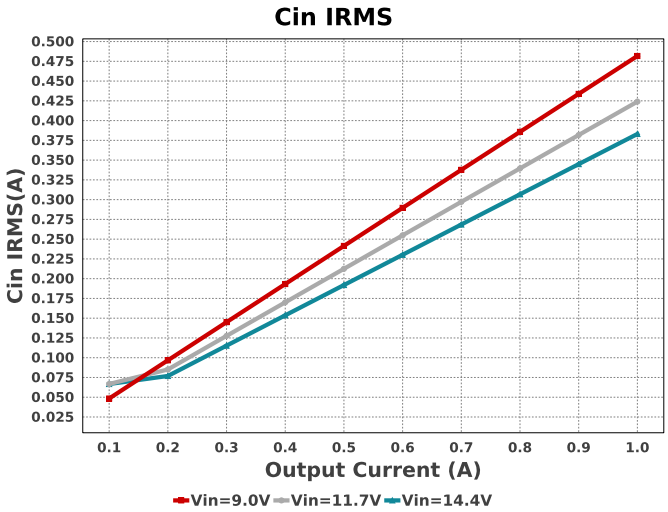

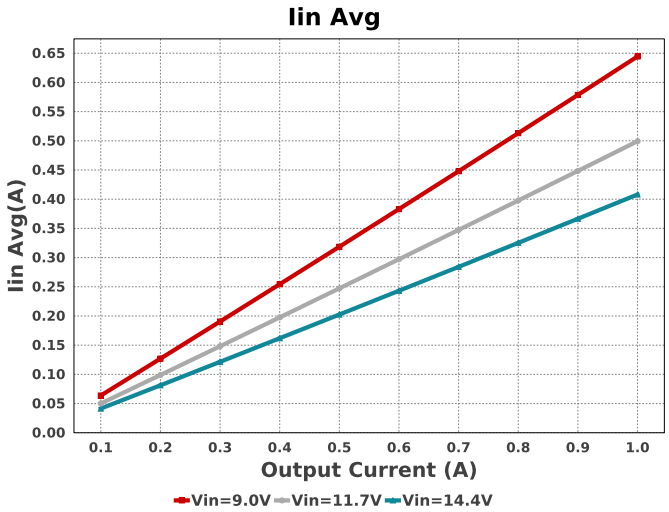

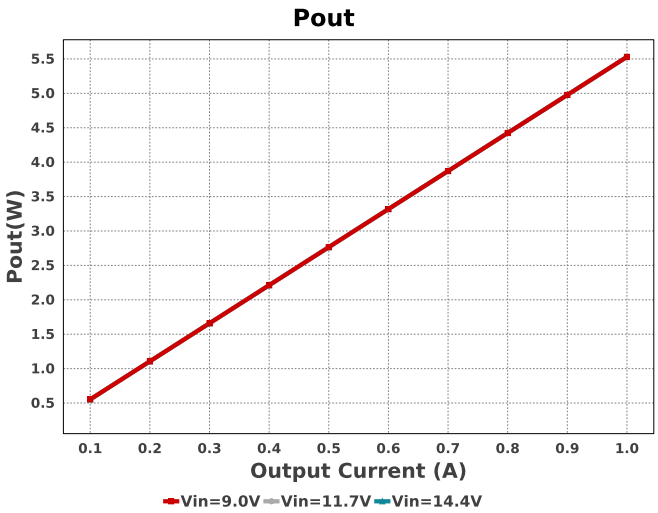

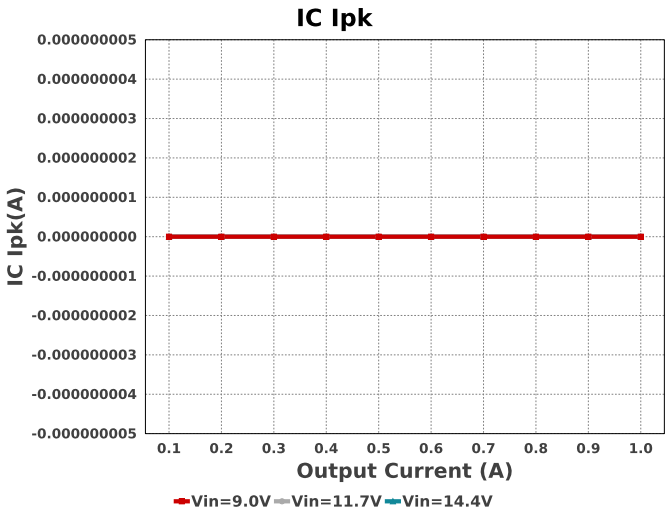

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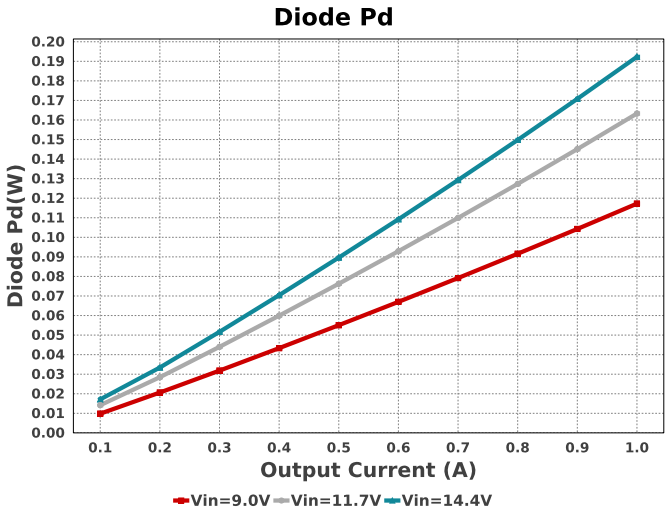
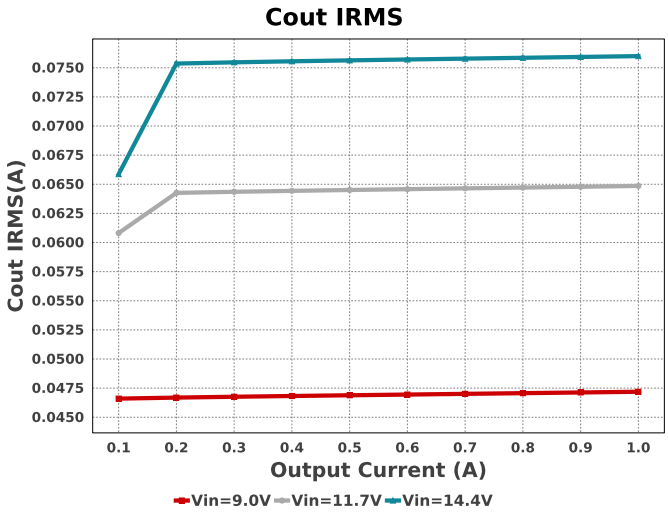
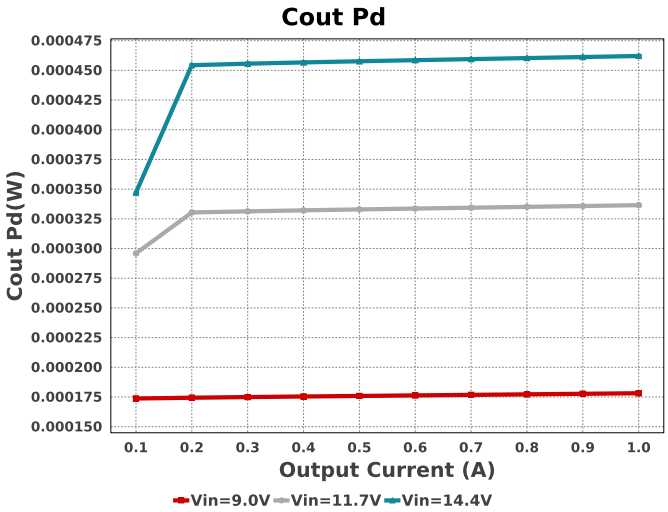
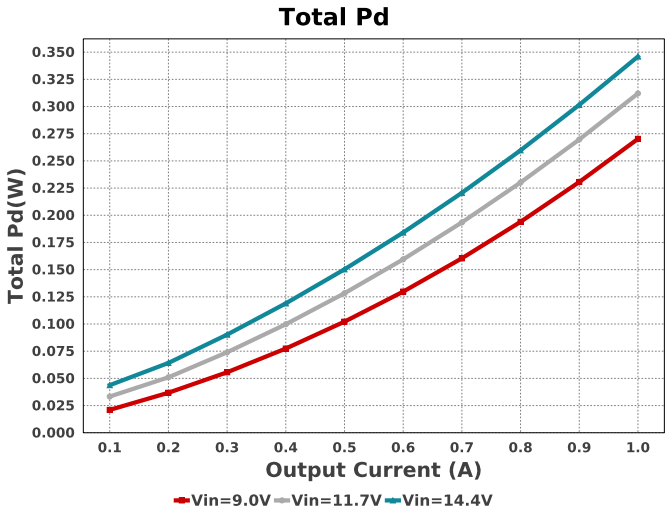
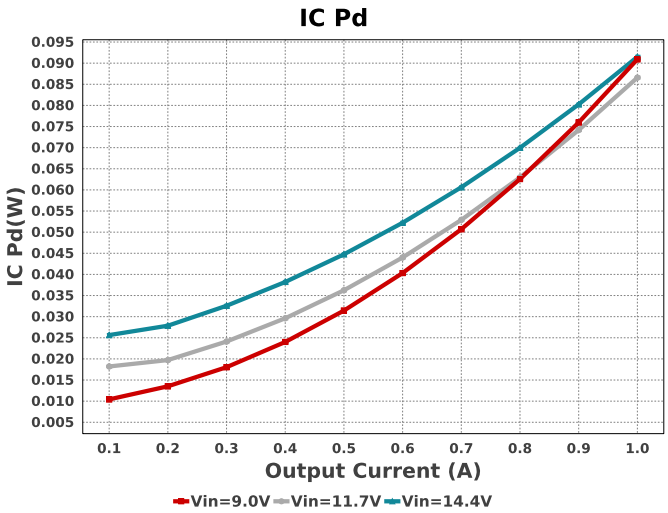
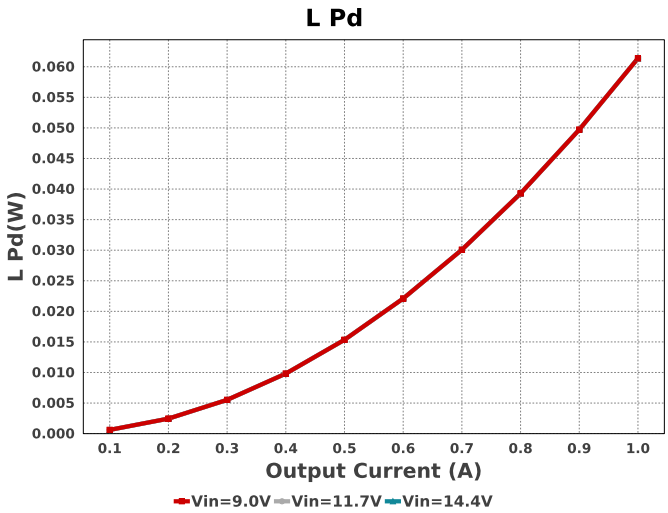

Diode Pd
Diode Pd(W)
0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.10 0.11 0.12 0.13 0.14 0.15 0.16 0.17 0.18 0.19 0.20
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V

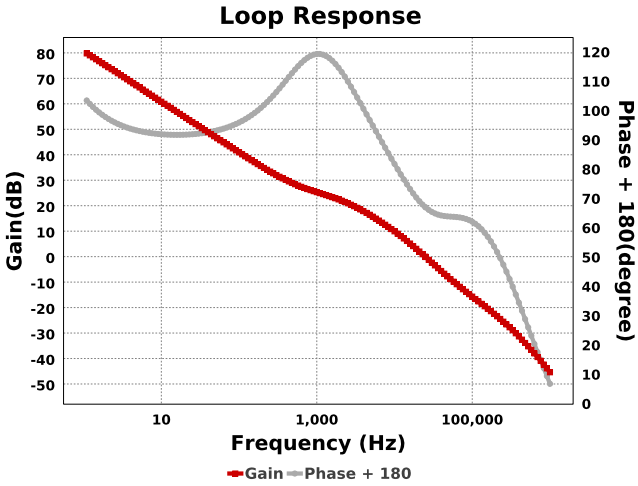

Cout IRMS
Cout IRMS(A)
0.0450 0.0475 0.0500 0.0525 0.0550 0.0575 0.0600 0.0625 0.0650 0.0675 0.0700 0.0725 0.0750
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V


Cout Pd
Cout Pd(W)
0.000150 0.000175 0.000200 0.000225 0.000250 0.000275 0.000300 0.000325 0.000350 0.000375 0.000400 0.000425 0.000450 0.000475
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V


Total Pd
Total Pd(W)
0.000 0.025 0.050 0.075 0.100 0.125 0.150 0.175 0.200 0.225 0.250 0.275 0.300 0.325 0.350
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V


IC Pd
IC Pd(W)
0.005 0.010 0.015 0.020 0.025 0.030 0.035 0.040 0.045 0.050 0.055 0.060 0.065 0.070 0.075 0.080 0.085 0.090 0.095
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V


L Pd
L Pd(W)
0.000 0.005 0.010 0.015 0.020 0.025 0.030 0.035 0.040 0.045 0.050 0.055 0.060
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
Output Current (A)
Vin=9.0V Vin=11.7V Vin=14.4V



Operating Values

#	Name	Value	Category	Description
1.	BOM Count	12		Total Design BOM count
2.	Total BOM	$3.829		Total BOM Cost
3.	Cin IRMS	383.061 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	344.24 µW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	76.002 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	462.1 µW	Capacitor	Output capacitor power dissipation
7.	Diode Pd	192.3 mW	Diode	Diode power dissipation
8.	IC Ipk	0.0 A	IC	Peak switch current in IC
9.	IC Pd	91.488 mW	IC	IC power dissipation
10.	IC Tj	62.379 degC	IC	IC junction temperature
11.	ICThetaJA Effective	26.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
12.	Iin Avg	408.05 mA	IC	Average input current
13.	L Ipp	263.277 mA	Inductor	Peak-to-peak inductor ripple current
14.	L Pd	61.38 mW	Inductor	Inductor power dissipation
15.	Cin Pd	344.24 µW	Power	Input capacitor power dissipation
16.	Cout Pd	462.1 µW	Power	Output capacitor power dissipation
17.	Diode Pd	192.3 mW	Power	Diode power dissipation
18.	IC Pd	91.488 mW	Power	IC power dissipation
19.	L Pd	61.38 mW	Power	Inductor power dissipation
20.	Total Pd	345.974 mW	Power	Total Power Dissipation
21.	Cross Freq	24.809 kHz	System Information	Bode plot crossover frequency
22.	Duty Cycle	39.939 %	System Information	Duty cycle
23.	Efficiency	94.112 %	System Information	Steady state efficiency
24.	FootPrint	310.0 mm^2	System Information	Total Foot Print Area of BOM components
25.	Frequency	611.622 kHz	System Information	Switching frequency
26.	Gain Marg	-48.924 dB	System Information	Bode Plot Gain Margin
27.	Iout	1.0 A	System Information	Iout operating point
28.	Low Freq Gain	79.781 dB	System Information	Gain at 1Hz
29.	Mode	CCM	System Information	Conduction Mode
30.	Phase Marg	67.268 deg	System Information	Bode Plot Phase Margin
31.	Pout	5.53 W	System Information	Total output power
32.	Vin	14.4 V	System Information	Vin operating point
33.	Vout	5.53 V	System Information	Operational Output Voltage
34.	Vout Actual	5.53 V	System Information	Vout Actual calculated based on selected voltage divider resistors
35.	Vout Tolerance	2.745 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
36.	Vout p-p	21.062 mV	System Information	Peak-to-peak output ripple voltage

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Design Inputs

Name	Value	Description
Iout	1.0	Maximum Output Current
VinMax	14.4	Maximum input voltage
VinMin	9.0	Minimum input voltage
Vout	5.53	Output Voltage
base_pn	TPS54560B-Q1	Base Product Number
source	DC	Input Source Type
Ta	60.0	Ambient temperature
UserFsw	600.0 k	Customer Selected Frequency

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

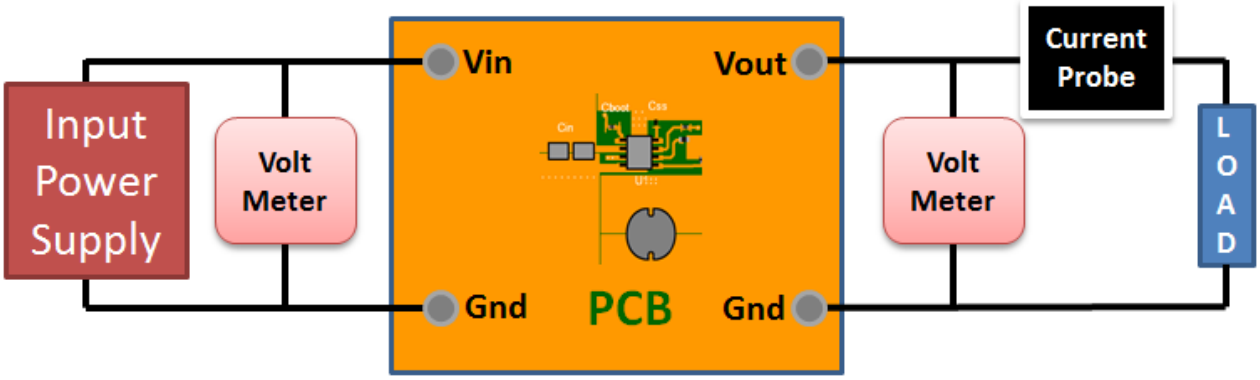
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 9.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

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2. Master key : 856C3ACFC3D186BD121B5CEC516967DA[v1]

3. **TPS54560B-Q1** Product Folder : http://www.ti.com/product/TPS54560B%2DQ1 : contains the data sheet and other resources.

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